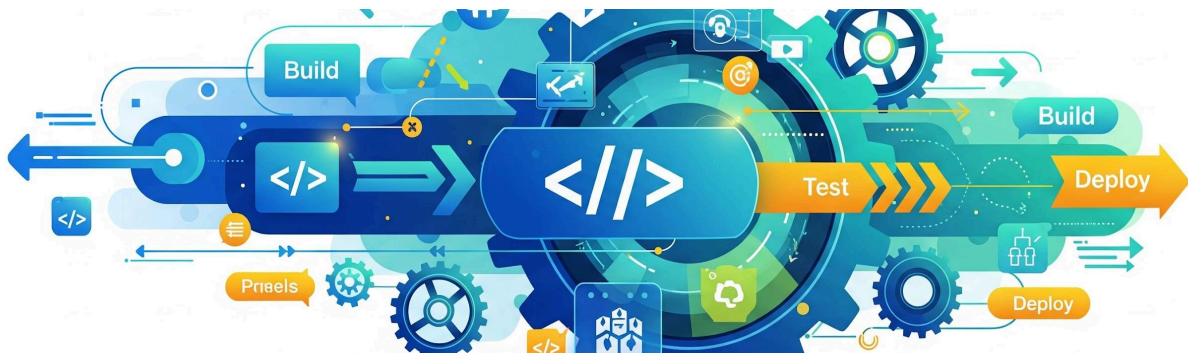




# Jenkins – Introduction (DevOps)

## What is Jenkins?

Jenkins is an automation tool used in DevOps to automate tasks such as building, testing, and deploying applications. Before Jenkins, developers had to run these tasks manually, which took time and caused errors.

Jenkins = Automation server for CI/CD - Continuous Integration/Continuous deployment

## Why Jenkins is Used

Jenkins helps automate the software delivery pipeline.

| Scenario        | Workflow                                      |
|-----------------|-----------------------------------------------|
| Without Jenkins | Developer → Manual Build → Manual Deployment  |
| With Jenkins    | Developer → GitHub → Jenkins → Build → Deploy |

Jenkins automatically:

- pulls code
- builds the project
- runs tests
- deploys the application

# Jenkins in DevOps Pipeline

Jenkins acts like the automation engine in DevOps.

## Example Pipeline:

Developer → GitHub → Jenkins → Build (Maven) → Deploy

## Key Features of Jenkins

- ✓ Open source
- ✓ Automates CI/CD pipelines
- ✓ Supports many plugins
- ✓ Integrates with Git, Docker, AWS, etc.
- ✓ Easy to extend

## What Jenkins Automates

Jenkins automates various tasks in the development workflow.

| Task # | Example Task                  |
|--------|-------------------------------|
| 1      | Pull code from GitHub         |
| 2      | Build application using Maven |
| 3      | Run automated tests           |
| 4      | Deploy application to server  |

## Jenkins Job

A Jenkins Job is a task that Jenkins performs.

## Example Job:

- Pull code from GitHub
- Build the project
- Deploy the application

# Jenkins Architecture

Jenkins runs on a server and executes automation tasks.

## Basic Components:

Developer → GitHub → Jenkins Server → Build / Deploy

## Real Example for Students

**Example scenario:** A developer pushes code to GitHub.

Jenkins automatically:

1. Detects the change
2. Pulls the new code
3. Builds the project
4. Deploys the application

No manual work needed.

## One Line Summary

Jenkins automates the process of building and deploying applications.

## Questions to Ask Students

1. What is Jenkins?
2. What problem does Jenkins solve?
3. Which stage of the DevOps pipeline uses Jenkins?

## Next Jenkins Class (Recommended)

The next class you should teach is **Jenkins Installation on EC2**.

| Step | Action                  |
|------|-------------------------|
| 1    | Launch EC2, Install Git |

| Step | Action                   |
|------|--------------------------|
| 2    | Install Java             |
| 3    | Install Jenkins          |
| 4    | Access Jenkins dashboard |
| 5    | Create first job         |

Installation :

```
sudo yum install java-21-amazon-corretto -y
```

```
java -version
```

```
sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/rpm-stable/jenkins.repo
```

```
sudo yum upgrade
```

```
sudo yum install jenkins
```

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable jenkins
```

```
sudo systemctl start jenkins
```

```
sudo systemctl status jenkins
```

## Jenkins Freestyle Job – Step by Step



Create a simple job that runs a command in Jenkins.

---

## ◆ **STEP 1: Open Jenkins**

Open in browser:

http://<EC2-Public-IP>:8080

Login to Jenkins.

---

## ◆ **STEP 2: Create New Job**

1. Click **New Item**
2. Enter job name:

my-freestyle-job

3. Select:  
☒ **Freestyle Project**
  4. Click **OK**
- 

## ◆ **STEP 3: Configure Job**

You will see configuration page.

---

### ◆ **General (Optional)**

- You can add description:

My first Jenkins job

---

### ♦ **Source Code Management (Optional)**

If using GitHub:

1. Select **Git**
2. Enter repo URL:

`https://github.com/<your-repo>.git`

👉 If not using GitHub, skip this section.

---

### ♦ **Build Triggers (Skip for now)**

Leave default.

(You can explain later: webhook, polling)

---

### ♦ **Build Step (Important)**

1. Scroll to **Build**
2. Click:  
➡ **Add build step**
3. Select:  
➡ **Execute shell**
4. Add command:

`echo "Hello from Jenkins Freestyle Job"`

---

## ♦ **STEP 4: Save Job**

Click:

 **Save**

---

## ◆ **STEP 5: Run Job**

1. Click:  
➡ **Build Now**
- 

## ◆ **STEP 6: Check Output**

1. Click build number (left side)
2. Click:  
➡ **Console Output**

You should see:

Hello from Jenkins Freestyle Job

 Job executed successfully!

---

## **Simple Explanation**

"Freestyle job is a simple Jenkins job where we manually define steps like running commands or building code."

---

## **Flow to Explain**

Jenkins → Job → Build → Output

